

even if these phones never hit the market. Concept designs help to peep into the future and are an art in the field of technology.

Handheld 3D laser scanners

Laser scanners are the best choice as they work with great precision with parts and shapes that are complex. As the technology advances, 3D laser scanners will be able to provide invaluable accuracy, resolution and integration at an unimaginable price.

There are a number of innovative handheld scanners available today which give you the benefit of portability to scan the deepest and remotest structure with great accuracy.

Dr Vaidya comments, "Portable laser scanner is one of the forms of new technological changes in 3D scanning as it gives accurate dimensions of an object directly to the machine. Portability allows one to take the 3D scanner anywhere and scan virtually anything, anywhere."

Advancements in semiconductors have made high-speed digitisers easily available. Many engineers believe that next-generation laser scanners will be the most notable and cutting-edge with the ability to recognise even the live features and model the scanned three-dimensional movements.

Just like cameras get their new features, scanners also undergo developments. Portable 3D scanners have innovative positioning targets and easy set-up for self-positioning. Because of the true portable design of the scanner, there is no limitation on the orientation of the scanner or accessibility in restricted spaces.

Structured light for 3D optical measurement

Structured light is one of the most commonly used techniques to obtain 3D optical measurements. The goal of a 3D optical measurement system is to capture the shape of an object in order to construct a three-dimensional model

that can be used for measurements and analysis. Applications range from machine vision systems that perform volumetric inspection in an assembly line to biometric systems that perform 3D facial recognition.

Structured light scanning is a variant of triangulation scanning where a pattern of laser stripes is projected on the object being scanned. When the laser line is swept over the solid object, the cameras are used to examine the deformation of the laser line pattern. The primary advantage of structured light scanners is speed, as they can scan multiple points or the entire field of view at a glance.

NextEngine, one of the leading manufacturers of laser scanners, has developed its own multistripe laser triangulation technology that uses four twin arrays of Class 1M solid-state lasers and two CMOS sensor camera modules. The eight lasers are managed by two Microchip microcontrollers and two quad-output operational amplifiers from Texas Instruments. The dedicated laser-control printed-circuit board is mounted to the laser assembly and connected to each laser via a three-lead flex connector.

Ganesh S., a business development manager at Texas Instruments, shares, "DLP-based structured light systems speed up the measurement process by using a digital projector to display a known pattern onto the object and a camera to capture the distortion of the pattern as it reflects off the object. These systems are extremely accurate, quick and cost-effective. The resulting data offers a wealth of valuable insights not easily available today."

Open Source 3D scanner

Open Source has become a *mantra* in many protocols, standards and software. Following the trend, MakerBot has launched the do-it-yourself version of MakerScanner—an Open Source 3D scanner.

The MakerScanner uses the tech-

nique of projecting energy into a scene in order to perform range-finding. The basics are similar to stereo vision. A laser pointer (scanning to make a vertical line) and a camera are offset at a certain base distance. As objects approach the sensor, the laser line appears closer to the edge of the camera's image. With correct calibration and some math, the range to each point can be calculated.

The basic hardware and tools needed to build this Open Source scanner include 3D printer, PS3 Eye USB camera, laser line pointer, two AA-size batteries, battery holder and Maker-Scanner software.

The Open Source 3D scanner from MakerBot is cheap, fast and accurate. The major cost involved is the camera. By using the right type of camera, surprisingly accurate 3D scanning can be achieved at a very low cost.

Does laser scanning fit your application?

There are several laser scanning technologies, but the main concern is which technology best suits your application. Laser scanning is used to scan large areas like terrestrial plains and buildings, to very small electronic components and devices. Some of the laser scanning technologies include confocal laser scanner, triangulation coordinate measuring machine scanner, portable 3D triangulation scanner, etc.

The QC Group suggests the use of a confocal laser scanner for very small objects. This scanner is capable of mapping topography to less than one micron on 3D position. It uses a precision confocal sensor with 0.01-micron resolution. Micro-scanning is used to capture the topography, which also needs to be scaled to a larger size without loss of detail. As in the case of embedded electronics, micro-scanning is also useful for mapping the characteristics of implantable devices or machine components. ●

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